Design and Analysis of Algorithms

Department of Computer Science & Engineering

DESIGN AND ANALYSIS OF ALGORITHMS

Dynamic Programming- Knapsack Problem

Department of Computer Science & Engineering

The Knapsack Problem

Problem Definition

- Given
  - n items of $\begin{array}{lllll} \text{integer weights :} & w_1 & w_2 & \dots & w_n \\ \text{values :} & v_1 & v_2 & \dots & v_n \end{array}$
  - knapsack of capacity W (integer $W > 0$)
- Find the most valuable subset of items such that sum of their weights does not exceed W

Kanpsack Recurrence

- To design a dynamic programming algorithm, we need to derive a recurrence relation that expresses a solution to an instance of the knapsack problem in terms of solutions to its smaller subinstances
- Consider the smaller knapsack problem where number of items is i ($i \leq n$) and the knapsack capacity id j ($j \leq W$)
- Then

$$F(i,j) = \begin{cases} \max(F(i-1,j),\ \ v_i + F(i-1,j-w_i)) & \text{if } j - w_i \geq 0 \\ F(i-1,j) & \text{if } j - w_i < 0 \end{cases}$$

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Dynamic Programming Example

item i	weight w_i	value v_i
1	2	12
2	1	10
3	3	20
4	2	15

What is the maximum value that can be stored in a knapsack of capacity 6?

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item i	weight w_i	value v_i
1	2	12

What is the maximum value that can be stored in a knapsack of capacity 6?

	capacity j				
i	1	2	3	4	5
1					
2					
3					
4					

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1	0	12	12		
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	capacity j				
i	1	2	3	4	5
1	0	12	12	12	12
2	10	12	22		
3					
4					

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3	10	12			
4					

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3	10	12	22	30	
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4					

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4	10				

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4	10	15	25	30	

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3	10	12	22	30	32
4	10	15	25	30	37

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2	10	12	22	22	22
3	10	12	22	30	32
4	10	15	25	30	37

Given above 6 items, maximum value that can be stored in a knapsack of capacity 5 is **37**

Complexity

- Space complexity: $\Theta(nW)$
- Time complexity: $\Theta(nW)$
- Time to compose optimal solution: $O(n)$